```python
from pgmpy.models import DiscreteBayesianNetwork
from pgmpy.factors.discrete import TabularCPD
from pgmpy.inference import VariableElimination

# Create Bayesian Network
model = DiscreteBayesianNetwork([
    ('Flu', 'Fever'),
    ('Flu', 'Cough'),
    ('Fever', 'Hospital'),
    ('Cough', 'Hospital')
])

# Prior Probability
cpd_flu = TabularCPD(
    variable='Flu',
    variable_card=2,
    values=[[0.9],
            [0.1]]
)

# Fever depends on Flu
cpd_fever = TabularCPD(
    variable='Fever',
    variable_card=2,
    values=[
        [0.8, 0.1],
        [0.2, 0.9]
    ],
    evidence=['Flu'],
    evidence_card=[2]
)

# Cough depends on Flu
cpd_cough = TabularCPD(
    variable='Cough',
    variable_card=2,
    values=[
        [0.7, 0.2],
        [0.3, 0.8]
    ],
    evidence=['Flu'],
    evidence_card=[2]
)

# Hospital depends on Fever and Cough
cpd_hospital = TabularCPD(
```

```python
# Hospital depends on Fever and Cough
cpd_hospital = TabularCPD(
    variable='Hospital',
    variable_card=2,
    values=[
        [0.90,0.30,0.20,0.05],
        [0.10,0.70,0.80,0.95]
    ],
    evidence=['Fever','Cough'],
    evidence_card=[2,2]
)

model.add_cpds(cpd_flu, cpd_fever, cpd_cough, cpd_hospital)

print("Model Valid:", model.check_model())

inference = VariableElimination(model)

print("\nCase 1 : No Evidence")
print(inference.query(variables=['Flu']))

print("\nCase 2 : Fever=True")
print(inference.query(
    variables=['Flu'],
    evidence={'Fever':1}
))

print("\nCase 3 : Fever=True, Cough=True")
print(inference.query(
    variables=['Flu'],
    evidence={'Fever':1,'Cough':1}
))

print("\nCase 4 : Hospital=True")
print(inference.query(
    variables=['Flu'],
    evidence={'Hospital':1}
))
```

```
/usr/local/lib/python3.12/dist-packages/pgmpy/estimators/__init__.py:4: FutureWarning: `pgmpy.estimators.StructureScore` is deprecated and will be removed in v1.3.0. Use `pgmpy.structure_score` instead
  from .StructureScore import (
Model Valid: True

Case 1 : No Evidence
+--------+------------+
| Flu    |   phi(Flu) |
+========+============+
| Flu(0) |     0.9000 |
+--------+------------+
| Flu(1) |     0.1000 |
+--------+------------+

Case 2 : Fever=True
+--------+------------+
| Flu    |   phi(Flu) |
+========+============+
| Flu(0) |     0.6667 |
+--------+------------+
| Flu(1) |     0.3333 |
+--------+------------+

Case 3 : Fever=True, Cough=True
+--------+------------+
| Flu    |   phi(Flu) |
+========+============+
| Flu(0) |     0.4286 |
+--------+------------+
| Flu(1) |     0.5714 |
+--------+------------+

Case 4 : Hospital=True
+--------+------------+
| Flu    |   phi(Flu) |
+========+============+
| Flu(0) |     0.7997 |
+--------+------------+
| Flu(1) |     0.2003 |
+--------+------------+
```